Neural Bellman Operators
Author Response to the Advisory Referee

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1 Report identity, complete author baseline and present scope

This is the next author revision of *Neural Bellman Operators* in its original recursive economic decision problem, with Econometrica numerical methods as the intended direction. This revision starts from the complete R119 author source `6250abfd317f3b2a56cb1303eb5ce78249bf4042`, published through `82618f037540233dda7d1ba8bfd3ddfdc88afbf9`. R110 remains its verified scientific ancestor; the completed R119 manuscript is not replaced by an earlier baseline or by a larger version label.

The initial complete remote observation at 02:07 UTC on 19 October 2026 found 360 heads: all 359 historical heads unchanged and the R119 publication branch added. The independent refresh at 07:48 UTC found all 360 of those heads unchanged and exactly one addition, the new R120 scientific-source branch, for 361 heads. The 36 review refs still contain 35 distinct persisted documents, including the R50 threshold screen; the R16 ref contributes no new report. The controlling advisory remains report commit `9723c6cd2cc8b3e071422f3e522d4a843946db63`, report blob `33ba8b144e2f7816b19b036404341402d23a582f`, assessing R97 manuscript `f802fe64d91d9f5c1c8f86123b0fbe114e622279`. Its complete bytes and all 35 retained report identities were reverified after the pause. It is an owner-commissioned AI-assisted advisory, not an Econometric Society decision and not a review of R119 or R120.

We have read the complete current main text, response and controlling report, verified all 35 retained report identities, and matched all 1,356 tracked R119 files (402,578,260 bytes) to their immutable publication-tree blobs. The source/PDF manifest and every R119 result remain unchanged. The correct retained active-tail proof path is `../2026-10-18-r119/new119/active_tail_proofs.tex`.

What has already been established. R119 proves a deterministic exact-point, univariate uniform-law AVaR width order $KR(1-\alpha)N^{-2}$ with risk-level-uniform constants for convex Lipschitz continuations and $N \geq 32$. It proves an outward original-policy action certificate, a bounded preliminary information budget and the soundness of packed evaluation. Its completed A/B/C service experiments retain every adverse phase and conventional control. The packed implementation wins ten of sixteen contemporaneous median-clock cells against geometric qualification, with six losses; all within-arm service variants nevertheless return the same complete policies. Independent validation D covers all 983 delivered policy keys and confirms the adverse ReLU-minus-quadratic mean-cost intervals and exact ReLU/ordinary-hinge identity. These are achieved R119 results, not fresh R120 observations.

The unresolved mechanism driving this revision. The new research target is an original-objective decision improvement with a quantitative mathematical and computational account. A tighter certificate for the same action does not supply such a gain. A strict comparison between completed policy brackets is already available in the inherited paper; a further contribution must identify how a candidate and its paid accuracy produce certified descent, convergence or justified stopping. If the training loss remains least squares, a successful policy correction still does not complete the separate action-aware acquisition request.

New mathematical work now supplied. Theorems 9 and 10 prove an original constrained-AVaR continuation modulus and an action-dependent integral certificate. For a fixed known tail, the comparison width has the matching information order $\min\{KR(1-\alpha)N^{-2}, ER(1-\alpha)N^{-1}\}$ under the explicit scalar point-query convention (Proposition S.8). Theorem S.9 bounds common-tail mismatch, and Propositions 11, S.10 and S.11 state the safe gain account, complete positive-margin refinement condition and sharp remaining-future information floor. The common-risk identity and projected support proposal rules retain their earlier ownership.

Proposition S.12 adds an independent rational all-action optimum witness, extending fixed-policy validation to original regret and action-distance bounds. Proposition 12 proves the fixed simultaneous transfer of certified gains to the declared finite initial-state population. Their mathematical validity and the separately executed economic findings are documented as distinct claims. The complete statements and proofs are in `new120/policy_improvement_main.tex` and `new120/policy_improvement_proofs.tex`.

Prospectively fixed execution design. The two remotely frozen protocols specified eight inherited representation arms and four complete services: the completed incumbent, unchanged R108 legacy-safe refinement, the new paired comparison with the same three support-gradient proposal rules, and the stronger direct original 512-node/24-bisection proposal using the same paid pair certificate. The source, actual actor and independent sample barriers preceded the completed 4,096 primary decisions and 6,144 descriptive timing decisions. The 64 new states are exact 40-bit dyadic draws, conditional on the declared independent uniform entropy assumption. All initial-state services are singletons; timing repetitions add no population observations. The ordinary original-V_1 value-label fitting objective is unchanged.

Actually published scientific source. Source barrier F is `369d02724cf12ae0e54390b4e7b5f894f703892e`, with tree `16c38ff041a7fa3c8972109e361494d4e5855c16` and the unchanged final R119 publication as its parent. The Git commit is dated 07:43:02 UTC on 19 October 2026; the actual remote publication was recorded at 07:45:41 UTC and independently observed at 07:48 UTC. The manifest's earlier 02:49 preparation time is not a remote publication time. The independent read-back matches the exact manifest and all 157 scientific inputs, totaling 1,206,453 bytes, to their SHA-256 values and the actual untruncated remote Git tree. The immutable manifest is `SOURCE_FREEZE120.json`; the actual publication receipt is `provenance/SOURCE_PUBLICATION120.json`, and the independent source/report check is `provenance/resumption120/20261019T074800Z/LINEAGE_SOURCE_PUBLICATION_VERIFIED120.json`. The source barrier fixes the scientific study; each subsequent barrier retains its own actual artifact, publication and execution identity.

Actually completed actor, sample and decision stages. The original value-label acquisition and all sixteen model files completed with exit zero under F. Their actual actor commit is `1c5e4dc7e04c108fe698d4c247a144b625db301c`, with publication receipt `provenance/ACTOR_P`

UBLICATION120.json. Only after this actor barrier did the program acquire the original 640 entropy bytes and construct the 64 two-coordinate, 40-bit states. The sample completed with exit zero and was published as aebc7a0175c0729b9ff5e833a7facc1ed5c4c2fd; its receipt is provenance/STATE_PUBLICATION120.json. The original acquisition and sample launch receipts remain in results120/launches/acquisition-attempt01/LAUNCH120.json and results120/launches/states-attempt01/LAUNCH120.json.

The fixed reporting specification was then published as 594d9336d89bedfd360e942146ae490e0fabd3a0. The primary launcher records this exact HEAD at 07:55:56.728551 UTC on 19 October 2026. The separate reporting-publication transport receipt was written at 08:00:26 UTC, after launch. It preserves the actual successful publication-before-launch operation sequence; its later recording time and Git committer date are not reported as independently observed prelaunch push timestamps.

The primary process and all three prescribed timing processes completed with exit zero at 08:09:59.755563 UTC. They retain all 4,096 primary decisions and 6,144 descriptive timing decisions. The actual outer primary-and-timing launch elapsed time is 843.024816 seconds, including its declared nested stages; it must not be added to its child timings as if those were disjoint costs. The full original receipt and stdout are bound by results120/launches/primary-attempt01/LAUNCH120.json. Execution completion alone does not supply an independently validated gain, a population confidence statement or a neural adoption region.

Completed scientific evidence. The campaign, independent validation, population analysis and first read-only reporter have now completed. All 4,096 primary decisions qualify; all 6,144 timing actions, intervals and recorded query counts match their corresponding primary rules. The separate rational calculation covers all 861 unique policy keys, 111,069 terminal contexts and 4,096 comparisons, including original all-action optimum bounds. The frozen 48-group, 192-event family gives 42 positive learned-incumbent event-staircase mean-gain lower bounds with its stated simultaneous 95% coverage. These are fresh R120 observations under the committed sample and fixed actors, not inherited R119 results.

We use C for the completed incumbent, L for inherited legacy-safe refinement, P for the new paired service and D for direct refinement. The substantive comparison is mixed. All learned own-incumbent gains are positive, but P has nine exact cell identities and seven unresolved cell-mean differences against L, and is slower in all 48 matched clocks. D is faster than P in all those clocks and has a strictly better original mean in five cells. The mixed bound is necessary at one actual Poly3 candidate, recorded once in each of P and D; the transferred incumbent deficit is inactive in all 2,290 acquired density pairs. All 512 ReLU–hinge policies are identical and every learned completed mean is strictly worse than the online control. Sections 10–10.2 and 10.3–10.5 develop these outcomes with all costs and remaining scientific boundaries.

Numerical decimals in this response summarize the retained exact rational endpoints; those endpoints determine every sign and price domain. The 992 same-sample cross-cell contrasts are not added to the prospectively declared population inference family. All older negative results, reserved studies and failed attempts remain separately identified.

2 Point-by-point response to the controlling advisory

The exact original requests and current source routes are preserved in provenance/R97_REQUIREMENTS120.json. The mathematical responses below identify work now written and proved. The scientific actions were fixed before sampling; the completed execution stages are identified above.

The independently completed validation and reporting stages now supply the actual economic contrasts and population findings below.

2.1 B1. Neural specificity remains unestablished

Original advisory request.

> Validity, recursive qualification, the AVaR enclosure, the information lower bound, and the fallback are representation-neutral. The ReLU family is not the complete-work winner and is never certified adopted over quadratic least squares.

Completed inherited response. The shared information theorem and original qualification remain representation-neutral. Exact ReLU/hinge identity now covers 96 paired files and 4,416 decisions. Fused qualification improves serving for multiple representations, but quadratic retains lower complete cost and original mean economic loss than ReLU. The separately frozen D validation adds independent original-primitive value bounds for all current delivered policies and retains all 56 arm comparisons. With the fixed left-minus-right arm ordering, the 56 contrasts comprise 32 strictly negative, 20 strictly positive, 2 exact-zero and 2 unresolved intervals. The two unresolved comparisons are quadratic versus the degree-two polynomial, one at each risk; the two exact zeros are ReLU versus the equivalent conventional hinge. The ReLU/ordinary-hinge equality and unchanged complete policies are explicit identities, so this validation supplies no architecture-exclusive economic gain.

Concrete R120 author action. The completed study gives every representation the same four services and original-model information. The ordinary hinge receives the ReLU features and coefficients and pays the full learned-feature acquisition charge. All 512 ReLU–hinge action pairs are exactly equal. All 42 learned own-incumbent cells have strictly positive independent mean-gain lower bounds, but every one of the 14 learned completed cells has a strictly worse sample mean original objective than the online semi-analytic control. The paired service P is slower than inherited L in all 48 matched clocks. Every one of the 32 declared ReLU/full-conventional-menu price intersections across all three clocks is empty, including the intersections of upper not-ruled-out conditions. Thus the achieved contribution is an action-relevant Bellman comparison and original-policy correction; this equal-feature experiment supplies no neural-specific policy or adoption advantage.

The completed evidence for this item is in Table II, Table III, Table B.VIII; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition S.8. The exact current file routes are `PROTOCOL120.md`, `code120/campaign120.py`, `code120/improvement120.py`, `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`.

Scientific boundary. A benefit from a common original-risk correction is a contribution to the NBO decision mechanism. It is not a neural-exclusive representation advantage. Any stronger neural claim requires a separate equal-information and equal-resource separation.

2.2 B2. The information-complexity result is too narrow for the platform claim

Original advisory request.

> The theorem is a specialized deterministic one-dimensional oracle result. It does not establish randomized, multivariate, nonuniform-law, or end-to-end Bellman complexity lower bounds.

Completed inherited response. The inherited information result includes an implementable algorithm that pays for tail localization and matches the lower order $KR(1-\alpha)N^{-2}$ under one shared deterministic exact-point convention. The original investment certificate carries this result into a feasible original-action calculation and pays its finite-precision error.

Concrete R120 author action. The new constrained-AVaR theorem supplies the original continuation gradient modulus and the action-distance modulus E. The fixed-known-tail information result has matching width order

$$\min\{KR(1-\alpha)N^{-2},\ ER(1-\alpha)N^{-1}\}.$$

One exact scalar value is one query; the claim is deterministic and univariate. The original smooth model also gives the stronger separate-width coefficient $2M_\alpha R^2(1-\alpha)^2$. The nonsmooth lower-bound adversary therefore does not establish an original-primitives or smooth-class lower bound. Full policy comparisons pay both risk deficits, outward point uncertainty and the unchanged future allowance. In execution the mixed upper is necessary at two recorded accepted levels for one common Poly3 candidate at $m = 4$; no wall-time gain or optimal general Bellman information complexity is inferred from that event. The full positive-margin refinement and future-information-floor propositions state when the interface can terminate successfully.

The completed evidence for this item is in Table B.III; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition 11, Proposition S.8, Theorem S.9, Proposition S.10, Proposition S.11. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`.

Scientific boundary. Randomized information bounds, nonuniform and multivariate risks, and full learned Bellman complexity require further results. The current scalar theorem keeps the cost of acquiring accurate continuation values explicit.

2.3 B3. The representation result is not decision-theoretic

Original advisory request.

> The ridge example separates restricted approximation classes but does not separate policies, service work, or economic outcomes. A conventional method with the same hinge is equivalent.

Completed inherited response. The representation comparison now includes a separately coded conventional oblique hinge carrying exactly the learned ReLU coefficients. Identical states/actions, certificates, rejected trials and complete policies are verified; differing clocks are attributed to implementation rather than neural expressiveness.

Concrete R120 author action. The new comparison authorizes changes of the original investment policy. It treats a difference of convex continuations as Lipschitz, without assuming that the difference is convex. The independent original-policy and all-action-optimum witnesses now provide value, gain, regret and action-distance bounds for every returned decision. For ReLU at $\alpha = 1/2$, P's mean original-policy gain over its own completed incumbent is approximately $[212.675, 213.885] \times 10^{-6}$; the mean-regret upper bound falls from 215.906×10^{-6} to 2.626×10^{-6}. The RMS action-error upper bound falls from 0.01470 to 0.001621. L returns the same ReLU policies, and conventional hinge is identical as well. These are original-investment improvements with a shared correction; they neither establish a neural geometry advantage nor borrow economic improvements from the separate posted-price example.

The completed evidence for this item is in Table II; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition S.8, Proposition S.12. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `INDEPENDENT_PROTOCOL120.md`, `code120/independent120.py`.

Scientific boundary. A decision-changing shared correction does not prove that a neural learner discovers unknown geometry more efficiently. Results in the distinct posted-price economy cannot be reported as investment improvements.

2.4 B4. The certificate-aware objective fails its intended service connection

Original advisory request.

> It improves residual span in 36 of 36 cells but worsens aggregate rejections and most quadratic cells. The missing theorem is the connection from the trained objective to action-relevant resolution.

Completed inherited response. We preserve all 36 favorable residual-span cells beside the increase from 644 to 757 rejections. New coupled-action, tail-defect and constrained-gap certificates target original action loss, while A/B/C empirically separate information reduction from actual decision-serving work. The inherited fitting objective itself is unchanged.

Concrete R120 author action. The acquired information now targets the original continuation difference at two feasible actions. A paid common-tail mismatch and a strict complete-policy gate connect this information to a permissible decision change, with the true future-controller allowance included once. Holding the three inherited support-gradient proposal rules fixed, P changes 391 of 1,024 actions and accepts 815 steps; L changes 388 and accepts 807. Their final actions differ in only 12 cases, and none of the 16 cell-mean comparisons resolves a strict incremental P advantage. The same paid pair primitive is supplied to the stronger direct conventional proposal. All fitted critics

still use the ordinary original-V_1 value-label objective. We therefore complete an action-dependent certificate and its original-policy evaluation, while retaining action-aware training and acquisition as a distinct unresolved scientific requirement.

The completed evidence for this item is in Table II, Table B.III; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition 11, Proposition S.8, Theorem S.9, Proposition S.10, Proposition S.11. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`, `code120/campaign120.py`.

Scientific boundary. This addresses the decision mechanism. If the acquisition remains the same least-squares fit, action-aware fitting itself is still unfinished. A genuine fitting-to-service claim additionally needs a specified action-aligned loss, an acquisition theorem and prospective validation of newly fitted actors.

2.5 B5. Complete work favors a strong conventional method

Original advisory request.

> Quadratic least squares is faster, uses fewer proposal queries, and has no neural acquisition overhead. R97 therefore does not shift the computational frontier toward NBO-specific representation.

Completed inherited response. The complete current tables charge actual labels, candidates, audits, serialization, loading, proposal work and all refused certificates. Packed qualification is competitive as a common method, yet current quadratic with its best measured conventional service remains cheaper and has lower original catalogue-mean cost than ReLU. The cost account now includes the full additional D verification bill: The actual external process cost is 1697.331508 seconds, including 1697.251692 seconds measured inside the process and 1693.603548 CPU seconds; recorded peak resident memory is 83,152 KiB, and compressed primitive traces occupy 48,026,132 bytes. The separately executed completed-record audit costs a further 7.705036 elapsed and 7.538375 CPU seconds. These independently implemented value enclosures strengthen the economic comparison but are not inserted into or substituted for any competitive service clock. All conventional victories and all unresolved independent contrasts remain visible.

Concrete R120 author action. The completed four-service/eight-arm comparison retains all 4,096 primary and 6,144 timing decisions and charges actual common-label acquisition, each standalone fit and audit, full hinge acquisition, loading, proposals, tail witnesses, precision ladders, rejected work, fallback and cohort bookkeeping. Summing complete standalone 64-state costs over the same 16 cells gives 94.478 seconds for L, 132.729 for P and 80.302 for direct refinement D. P purchases fewer compiled terminal rows than L, yet is slower in every primary cell and all 48 matched fresh clocks. D purchases substantially more terminal rows and is faster than P in all those clocks. The independent rational study costs a separate 1,806.153 seconds and remains charged as research validation. These complete scopes replace a point-count-only competitiveness claim; they retain the stronger traditional method and every adverse clock.

The completed evidence for this item is in Table III; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Proposition S.12. The exact current file routes are `PROTOCOL120.md`, `code120/campaign120.py`, `code120/improvement120.py`, `new120/policy_improvement_proofs.tex`, `INDEPENDENT_PROTOCOL120.md`, `code120/independent120.py`.

Scientific boundary. R119's conventional quadratic winner remains a historical fact. An improvement over a neural arm's own incumbent, or fewer calls in a shared algorithm, does not establish a new neural complete-cost frontier.

2.6 B6. The purchase exercise is conditional sensitivity, not economic identification

Original advisory request.

> No learned service is adopted over quadratic least squares. Prices and fees are external, clocks are descriptive, and the policy interval is broad. The exercise does not change an economic conclusion.

Completed inherited response. Current adoption is expressed as a sufficient inequality in actually observed acquisition/service components and an explicitly external economic price. All fifteen within-arm policy contrasts are exactly zero, and the quadratic/ReLU economic comparison is adverse rather than converted into a purchase success. D independently reports all 56 arm-mean cost intervals, and separately checks every one of the 240 within-arm service-pair identity cells. All 240 service-pair cells contain identical complete policies at every one of their 144 states, giving exact zero mean differences rather than an inference from overlapping intervals. This confirms the scope of the current policy comparison while adding no calibrated compute price, demand curve, fee or population adoption estimate.

Concrete R120 author action. The fixed 48-group, 192-event population calculation has actually completed on the prospectively drawn 64 states. Under its stated finite-uniform population and independent-entropy assumption, the simultaneous 95% family gives positive event-staircase mean-gain lower bounds in all 42 learned own-incumbent groups, ranging from 2.661×10^{-6} to 14.192×10^{-6}. The six online groups have zero observed gains and zero lower bounds; the ordinary bounded-mean lower bounds alone are zero. The reported price sets use actual paid work and the declared conventional menu. Positive own-incumbent price regions coexist with empty ReLU/full-menu regions. No calibrated monetary value, preference parameter, market demand or population adoption conclusion is supplied by those normalized-objective calculations.

The completed evidence for this item is in Table IV, Table B.VIII; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Proposition 11, Proposition 12, Theorem S.9, Proposition S.10, Proposition S.11. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`, `INDEPENDENT_PROTOCOL120.md`, `code120/stats120.py`.

Scientific boundary. Symbolic price thresholds and deterministic catalogue means do not identify calibrated economic adoption. The historical zero-adoption-over-quadratic result and all unresolved regions remain intact.

2.7 B7. The new 84 actors lack direct population policy-cost inference

Original advisory request.

> R97 transfers only the uniform deterministic qualification bound to the new services. The R96 statistical intervals belong to different frozen actors and cannot establish the new actors' mean-cost ordering.

Completed inherited response. The text now attributes every policy-cost interval to its exact current actor and original state catalogue. R119 has direct deterministic reference intervals for newly acquired actors, but neither them nor R96 statistical intervals are assigned to the 84 R97 actors. D provides independent original-primitive value enclosures for all 983 actual C keys and all 96 current arm/service/risk means. It was prospectively frozen after the known C results and is expressly not an unseen sample. No D policy, rational node or current deterministic mean is transferred to the 84 R97 actors or counted as independent population evidence.

Concrete R120 author action. The source F, actual actors M and single 640-byte state draw D are separately and actually committed before evaluation. The recorded 64 exact 40-bit dyadic states are shared across all arms and risks, and singleton initial-state services prevent a batch fallback from making one row's rule depend on another row. All 4,096 original-policy comparisons and all 48-group/192-event statistics completed under these commitments. The 95% guarantee uses the fixed one-sided binomial allocation and bounded-mean allocation from the frozen protocol; it conditions on the declared independent uniform entropy law and fixed actors. This supplies the new R120 population evidence. It neither completes an old R97 actor family nor imports R98/R115 actors, reserved states or inference into this experiment.

The completed evidence for this item is in Table IV; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Proposition 12. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `PROTOCOL120.md`, `INDEPENDENT_PROTOCOL120.md`, `code120/stats120.py`, `code120/custody120.py`, `code120/io120.py`, `code120/campaign120.py`.

Scientific boundary. Neither R96 intervals nor the independently committed R98 or R115 actors and samples belong to R120. A changed returned action also cannot reuse an old D value enclosure as though its policy key were unchanged.

2.8 B8. Independent verification stops before fitting

Original advisory request.

> Feature selection, interval-aware fitting, rounding, domain validation, and serialization are not independently reproduced. This is the critical new acquisition stack.

Completed inherited response. The current acquisition audit independently reconstructs feature matrices, enumerates feasible nonlinear active faces and uses pivoted QR to check every nonexcluded candidate objective and selection. Rounded coefficients and serialized reload predictions are checked, and every candidate/audit cost is charged. D adds a separate no-production-import

evaluation of every delivered C policy key from original economic primitives. Its scope begins after action selection; the matrix/QR fitting reconstruction remains the evidence for acquisition checks. D does not turn a value-validation run into a second fitting algorithm or an independent laboratory replication.

Concrete R120 author action. The separately implemented rational validator actually evaluates all 861 unique returned policy keys, with 111,069 terminal contexts and 4,096 complete-policy comparisons. Its true Lagrange affine supports also give an original all-action optimum lower witness, rather than attaching an unproved slope to a scalar lower value. Thus the independent calculation covers regret and action-distance bounds as well as policy value. All 4,096 intervals overlap: 726 production intervals contain the independent enclosure, while 3,370 narrower production intervals are contained in it. Only the former direction inherits coverage from the independent enclosure; overlap or the latter direction does not independently prove fine production endpoints. The validator does not import the production terminal compiler, retains its exact rational traces and 1,806.153-second cost, and claims neither external authorship nor a complete historical fitting-optimizer replication.

The exact completed source/evidence route includes `results120/independent-attempt01/EXECUTION120.json`; the current response map binds all corresponding files.

R120 source route. The written mathematical route is Proposition S.12. The exact current file routes are `new120/policy_improvement_proofs.tex`, `INDEPENDENT_PROTOCOL120.md`, `code120/independent120.py`, `PROTOCOL120.md`, `code120/custody120.py`, `code120/io120.py`, `code120/campaign120.py`.

Scientific boundary. Shared floating-point libraries or a reused terminal compiler must remain disclosed. Checking a stored fitted actor is not a replay of every historical fitting optimizer, and interval overlap alone does not prove a fine endpoint.

2.9 B9. The active submission remains cumulative and disproportionate

Original advisory request.

> A 47-page article, 37-page supplement, and 230-page appendix are excessive for the narrow current theorem and negative acquisition experiment. Historical preservation need not determine the active submission's length.

Completed inherited response. The author revision gives the current theory–algorithm–original-economy sequence its own main sections and complete proofs; every historical result, failed experiment and negative cell remains in a separately navigable development/retention structure.

Concrete R120 author action. The active main argument now follows the original continuation/actor problem, the action-dependent modulus, a usable complete-policy improvement rule, the executed matched comparison and its economic interpretation. The supplement contains every new proof and the independently implemented original-optimum construction. The development tables retain all 64 objective/cost cells, 192 clocks, 192 threshold events, 992 sample contrasts and every prescribed price comparison. The 35 original report bodies and the 1,049 inherited

item/parent routes remain accessible with exact R119 source bindings. Their presence does not replace the new evidence: the main text reports P's adverse timing, unresolved incremental economic advantage and the stronger conventional proposal alongside its valid gains and the one observed mixed-bound acceptance.

The completed evidence for this item is in Table II; the exact row files and hashes are retained in the current response map.

R120 source route. The exact current file routes are `CONTENT_RETENTION_MAP120.md`, `REVISION_BOUNDARY120.md`, `provenance/HISTORICAL_ROUTE_BINDINGS120.json`.

Scientific boundary. Document reorganization does not close a scientific concern. No credible theorem, proof, negative result or original application is removed to shorten the active presentation.

2.10 B10. The negative result is more compelling than the platform claim

Original advisory request.

> The strongest new fact is that a valid global residual objective can fail to improve delivered decisions. The manuscript should either make that misalignment central or supply a new action-aligned objective and theorem.

Completed inherited response. The failure of residual fitting to improve actions is made explicit and retained as a primary scientific finding. R119 advances action-valid qualification and paid risk acquisition; masking first fails in full time, and a separately frozen packed implementation then improves the same-information service without inventing policy gains.

Concrete R120 author action. The new method buys information about an original action contrast and can change the decision. The continuation modulus, fixed-tail information bounds and quantitative tail-defect/refinement analysis are proved; the completed-incumbent gate and the three support proposals retain their R108–R110 ownership. The completed diagnostics distinguish use from independent contribution: the mixed upper is active at 5,961 of 9,016 paid levels and necessary at two accepted levels representing one Poly3 candidate shared by P and D. The transferred incumbent tail-defect bound is inactive in all 2,290 acquired density pairs. P's final action differs from L in 12 cases, with nine exact cell identities and seven unresolved mean differences; its full cost is higher in all 16 primary cells. We claim the proved comparison and actual fixed-resolution decision example, while preserving the unresolved incremental competitiveness and the unfinished action-aware acquisition problem.

The completed evidence for this item is in Table B.III, Table III; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition 11, Proposition S.8, Theorem S.9, Proposition S.10, Proposition S.11. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`.

Scientific boundary. The R119 phases retain exact-zero within-arm policy gains. The executed R120 gains belong only to the new source and cannot retrospectively turn the R97 acquisition objective into successful action-aware learning.

2.11 M1. Recast the paper around one method-neutral question

Original advisory request.

> Ask when a stored continuation can be safely purchased for recursive-risk dynamic decisions. Treat neural networks as one optional representation, not the source of validity.

Completed inherited response. The manuscript retains Neural Bellman Operators as the original economic research identity and centers the numerical question on when learned or conventional stored continuations can deliver original recursive decisions at justified loss and complete cost. Validity, information and architecture effects are separately stated.

Concrete R120 author action. Neural Bellman Operators remains the same original-economic-decision author project. The main question is how learned continuations, recursive feasible actors and action-relevant information can produce valid improvements at justified complete cost. The investment model now supplies actual policy changes, original-value/regret bounds and fixed-law gain inference, rather than an internal fit metric alone. Its continuous shock changes stock imbalance, while investment trades current adjustment cost against the original risk-sensitive continuation and capacity constraint. The main interpretation also confronts the strong online and direct controls and full paid work. The new safeguard is shared across representations; those facts preserve the neural Bellman research identity without assigning an unsupported architecture-exclusive mechanism.

The completed evidence for this item is in Table II; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition S.8. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`, `code120/campaign120.py`.

Scientific boundary. The advisory's proposed change of title, venue or paper identity is not adopted. Continuing the research program is an author decision; the scientific objections still require the mathematical and numerical work listed here.

2.12 M2. Establish the true novelty of the information theorem

Original advisory request.

> Give a theorem-by-theorem comparison with information-based complexity. Add a genuinely new randomized, nonuniform-law, multivariate, or Bellman-compositional result, or present the current theorem as a specialized supporting lemma.

Completed inherited response. The new theorem isolates paid tail localization, proves a risk-level-uniform upper rate and compares it with the matching deterministic exact-point lower account. The independent 320-cell rational catalogue preserves all conventional wins and all actual query counts, separate from original-economy implementation.

Concrete R120 author action. The novelty statement distinguishes the new original constrained-AVaR smoothness calculation, action-distance integral interface, matching fixed-tail query order and quantitative constrained-tail transfer from the inherited common-risk identity and strict policy gate. The original model's additional smoothness gives its own stronger upper coefficient and is excluded from the generic nonsmooth lower-class claim. The full finite-resolution theorem includes paid deficits and the true future-policy floor. Actual execution shows one Poly3 acceptance at a paid resolution that the separate convex upper cannot authorize at that same resolution. It does not measure a traditional method after another refinement, a counterfactual time saving, or general end-to-end Bellman optimality. The economic and timing comparisons retain those precise limits.

The completed evidence for this item is in Table B.III; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition 11, Proposition 12, Proposition S.8, Theorem S.9, Proposition S.10, Proposition S.11. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`, `INDEPENDENT_PROTOCOL120.md`, `code120/stats120.py`.

Scientific boundary. An implementation of a known accept-if-better rule alone is not a new convergence theorem. Broader randomized or multivariate claims require their own lower/upper bounds and are not inferred from scalar evidence.

2.13 M3. Develop an action-aligned acquisition objective

Original advisory request.

> Construct a loss that controls rejection, endpoint work, or policy-cost resolution on a predeclared context law. Prove the connection and validate it prospectively against complete work.

Completed inherited response. R119 supplies certificates for action comparisons, continuation-error transport and original-policy all-action lower witnesses. It evaluates policy cost, rejections and complete work. The inherited least squares training objective remains unchanged.

R119 validation D independently evaluates the full delivered-policy catalogue and retains every direct economic contrast, including zero and unresolved intervals. It changes no labels, fitted coefficients, proposals or selection criterion. This value evidence therefore leaves the separate request for an action-aware training objective open.

Concrete R120 author action. All 16 stored model/risk artifacts were freshly acquired under the frozen ordinary original-V_1 interval-midpoint value-label rule, with 256 common sites per risk and complete fitting/audit costs. The new action-relevant acquisition occurs inside the paid policy-comparison interface; it does not change that training objective. Its original-policy consequences are independently evaluated, including 42 positive learned-incumbent gain groups and the unresolved P-versus-L incremental mean comparison. The item therefore separates the completed comparison-information theorem from the still-needed action-aligned loss, acquisition theorem and newly fitted-actor validation. R98's distinct seven-objective acquisition experiment remains preserved under its own unchanged actor and sample barriers.

The completed evidence for this item is in Table II, Table III; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition S.8. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`, `code120/campaign120.py`.

Scientific boundary. A policy correction can materially advance the original decision problem while leaving action-aware acquisition open. The response will not label unchanged least-squares training as a newly validated learning objective.

2.14 M4. Expand the conventional geometry frontier

Original advisory request.

> Compare ReLU with conventional oblique hinges, adaptive triangulation, convex regression, and representation-neutral basis selection under equal information, rank, acquisition, and validation budgets.

Completed inherited response. The current frontier includes exact conventional oblique hinges, adaptive fixed-dictionary nonlinear selection, full polynomials, grid interpolation and online semi-analytic control, with common labels and actual rank/byte/acquisition accounting. Hinge equivalence is tested at the delivered-policy and certificate level.

Concrete R120 author action. The contemporary frontier contains online semi-analytic, quadratic, ReLU, equal-feature hinge, spline, degree-two polynomial, degree-three polynomial and grid arms, each with C, L, P and D. All 992 unordered within-risk cell comparisons are retained. Every learned completed cell is strictly worse than online in sample original value, and all 512 ReLU–hinge policy pairs are identical under full equal acquisition charges. D has a strictly better independent mean than L and P in five cells, two exact online identities and nine unresolved comparisons; it is faster than P in every matched clock. Thus conventional geometry, direct original-formula access and the shared certificate all receive their actual competitors. A favorable comparison with only a neural arm's own incumbent is not used as the representation frontier.

The completed evidence for this item is in Table II, Table B.VIII; the exact row files and hashes are retained in the current response map.

R120 source route. The exact current file routes are `PROTOCOL120.md`, `code120/campaign120.py`, `code120/improvement120.py`.

Scientific boundary. A geometry supplied to ReLU and withheld from a comparator cannot support neural specificity. Unexecuted adaptive conventional classes remain explicit missing comparisons.

2.15 M5. Treat the negative result as first-class evidence

Original advisory request.

> Report aggregate and cell-level rejection changes beside span improvements, explain where the global objective allocates approximation effort, and do not label 36/36 span reduction as delivered-service success.

Completed inherited response. The retained R97 summary states 36/36 span reductions together with rejections 644 to 757, quadratic 91 to 210, ReLU 286 to 287 and spline 267 to 260. New tables retain every A/B/C loss, including B masking overhead and six C geometric median losses, without counting internal-metric reduction as economic success.

Concrete R120 author action. The main text and complete development tables retain P's 16 primary cost losses and all 48 fresh-clock losses to L, nine exact and seven unresolved P/L mean comparisons, and all six zero-gain online population groups. P retains 2,257 rejected candidates, including 785 exhausted precision ladders; D retains 633 rejections, including 194 exhausted ladders. The inactive transferred-deficit result and the distinction between 5,961 active mixed bounds and only one necessary economic candidate are reported directly. The older R97 residual-span mismatch, adverse R119 phases, conventional wins and unverified neural-adoption regions remain unchanged. No discarded clock, unresolved sign or failed attempt is removed to make the new mechanism appear uniformly favorable.

The completed evidence for this item is in Table B.III, Table III; the exact row files and hashes are retained in the current response map.

R120 source route. The exact current file routes are `PROTOCOL120.md`, `code120/campaign120.py`, `code120/improvement120.py`, `CONTENT_RETENTION_MAP120.md`, `REVISION_BOUNDARY120.md`, `provenance/HISTORICAL_ROUTE_BINDINGS120.json`.

Scientific boundary. No adverse result is replaced by a fresh timing run or a different tolerance. A successful scientific audit is not a favorable economic result.

2.16 M6. Execute a consequential application or narrow the claim

Original advisory request.

> Either calibrate policy margins, compute prices, fees, and workloads in an application where adoption matters, or state that the paper is about numerical certification rather than economic purchase.

Completed inherited response. We preserve the Econometrica goal and original economic model, report direct original cost and action bounds, and show where actual common-method information savings become complete-service savings. Compute prices, fees, workloads and normalized-loss conversion are explicitly uncalibrated. D strengthens the direct original-economic value evidence with an exhaustive second implementation on the current catalogue. With the fixed left-minus-right arm ordering, the 56 contrasts comprise 32 strictly negative, 20 strictly positive, 2 exact-zero and 2 unresolved intervals. The two unresolved comparisons are quadratic versus the degree-two polynomial, one at each risk; the two exact zeros are ReLU versus the equivalent conventional hinge. It remains a two-date investment validation, with no newly calibrated welfare unit, compute price, workload population or enlarged application attached to it.

Concrete R120 author action. The economic interpretation reports the original risk-sensitive investment objective, not a different loss or welfare surrogate. It shows how the independently bounded ReLU own-incumbent gains reduce original regret and action-distance ceilings, and why a larger spline gain can reflect a worse incumbent. The online semi-analytic policy is already strictly better than every learned completed policy on the sample, while direct refinement is a material traditional competitor. The simultaneous 42 positive learned-incumbent lower bounds and all actual price scenarios are reported with their population and cost scopes. The units remain normalized original-objective units: this two-date experiment adds decision-relevant evidence but does not claim monetary, consumption-equivalent or field-adoption calibration.

The completed evidence for this item is in Table II, Table IV; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Proposition 11, Proposition 12, Theorem S.9, Proposition S.10, Proposition S.11, Proposition S.12. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`, `INDEPENDENT_PROTOCOL120.md`, `code120/independent120.py`, `code120/stats120.py`.

Scientific boundary. A synthetic catalogue and a price-sensitivity inequality are not an empirically calibrated adoption decision. The goal of a substantive economic application remains active.

2.17 M7. Add a genuine scaling experiment

Original advisory request.

> Vary horizon, controls, innovation dimension, and risk level under matched comparators. Report product-point counts, memory, endpoint work, acquisition, and complete service work.

Completed inherited response. The current matched original-kernel study varies d=2,4,8 at both original risks and charges common proposal plus certificate work. Complete two-dimensional A/B/C services separately retain every operation/cost and the older adverse vector tree is preserved.

Concrete R120 author action. The completed experiment is explicitly two-dimensional, two-date and evaluated at the two prescribed risk levels. It reports each service's full acquisition, original-value queries, candidate/refinement counters, raw process costs and complete resource-vector frontier. The four predeclared amortization volumes are conditions computed from these retained measurements; they are not measured throughput, higher-dimensional scaling or a horizon experiment. The new information theorem has a scalar-tail scope, and the generic lower bound excludes richer original-formula access. Existing higher-dimensional and application evidence retains its own source and limitations. Broader scaling and consequential application claims remain concrete research obligations rather than being filled by these 64 new initial states.

The completed evidence for this item is in Table III, Table B.VIII; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition S.8. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `CONTENT_RETENTION_MAP120.md`, `REVISION_BOUNDARY120.md`, `provenance/HISTORICAL_ROUTE_BINDINGS120.json`.

Scientific boundary. The existing d=2,4,8 kernel checks do not supply learned high-dimensional or longer-horizon evidence. The univariate active-tail proof cannot be reused as a multivariate tractability proof.

2.18 M8. Strengthen timing evidence

Original advisory request.

> Use more repetitions, record or control processor state, and make operation counts primary. Predeclare the inferential target for deployment clocks if a probabilistic purchase claim is desired.

Completed inherited response. All seven fresh process clocks and actual primitive counters are reported; source/protocol fixes ordering and complete workloads. Other controlled computation and formatting are paused for B/C, while A overlap and all prior observations remain disclosed. No warmup or unfavorable repeat is discarded.

Concrete R120 author action. All three 32-state timing processes completed, retaining 6,144 timed actions and 192 cohort clocks. Every timed action, interval and recorded query count matches its corresponding primary rule; no mismatch or clock was censored. P is slower than L in all 48 matched clocks; D is faster than P in all 48 and faster than L in 41, with seven losses and two adverse cell medians. Cohort measurements, standalone acquisition charges, loading and larger process/launcher scopes are labeled separately. These are descriptive engineering observations under the observed execution conditions. The clocks are not extra population observations and do not supply confidence bounds for future hardware, latency, energy prices or external load.

The completed evidence for this item is in Table III; the exact row files and hashes are retained in the current response map.

R120 source route. The exact current file routes are `PROTOCOL120.md`, `code120/campaign120.py`, `code120/improvement120.py`.

Scientific boundary. Observed engineering clocks are not confidence bounds for future hardware, energy prices or a deployment population. A probabilistic latency claim needs a separately specified inferential target.

2.19 M9. Independently reimplement fitting and package a standalone environment

Original advisory request.

> Reproduce feature selection, interval-aware fitting, rounding, whole-domain validation, and serialization independently. Include a container or lockfile and all inherited modules needed to run the new tests from the artifact alone.

Completed inherited response. The acquired fitting stack now has independent matrix/active-face QR verification, candidate selection and rounded-payload checks. Frozen117/frozen108 dependency closure, actual scientific source hashes, numerical library versions, original-kernel tests and exact record auditors travel with the current revision. D is a separately source-bound standard-library implementation of the value-enclosure layer, importing only the unchanged Fraction economic primitive. It records exact delivered inputs, all terminal traces and full process cost. Its narrower environment boundary is explicit: it does not independently recreate the learned fitting stack or certify floating linear algebra exactly.

Concrete R120 author action. The exact source F, actual actor M, new sample D and pre-outcome reporting specification R are preserved together with their real launch and publication receipts. Primary evidence was separately published as E0 before final reporting. Independent original-primitives validation, the fixed rational statistics and the first read-only reporter have all completed with exit zero; the reporter binds 777 unchanged input files and 36 stable products. The derivative algebra diagnostic preserves its first source-path failure and the corrected second attempt, without changing frozen science. The bounded package audit and source CI distinguish record/byte checks and statistical/report replays from a fresh primitive-economy rerun. Actual final source, PDFs, build logs and CI states belong to the separate source/PDF and publication declarations; no R119 success or future R120 job is asserted here as a completed R120 CI.

The exact completed source/evidence route includes `provenance/SOURCE_PUBLICATION120.json`; the current response map binds all corresponding files.

R120 source route. The written mathematical route is Proposition S.12. The exact current file routes are `PROTOCOL120.md`, `INDEPENDENT_PROTOCOL120.md`, `code120/custody120.py`, `code120/io120.py`, `code120/campaign120.py`, `new120/policy_improvement_proofs.tex`, `code120/independent120.py`.

Scientific boundary. An existing R119 workflow success cannot certify an R120 build or experiment. The new source/PDF package and any CI claim need their own completed receipts; queued or merely repaired workflows are not successes.

2.20 M10. Shorten the active submission substantially

Original advisory request.

> Move historical material to an archival companion. The active article should contain only the current qualification theorem, precise information result, acquisition objective, matched experiment, and supported economic interpretation.

Completed inherited response. Current original Bellman assumptions, coupled-action qualification, active-tail information, implementation and the matched economic study are presented as one author chain. The full historical content and A/B/C raw frontiers remain in development companions and immutable provenance rather than flooding the main contribution.

Concrete R120 author action. The main paper concentrates on the original recursive decision, complete new mathematical argument, contemporaneous experiment and economic interpretation. Full assumptions, proofs, independent optimum verification and all detailed tables remain in the

supplement/development companion. All 35 exact original reports are copied unchanged into the current source package; the 620 major, 394 additional and 35 overlapping-parent treatments retain their original identities and specific remaining requirements. The current response supplies an actual proof/evidence route for each of the 30 controlling items. This organization preserves NBO's learned continuation and actor problem, the strong traditional comparisons and every valid historical result, while keeping the current contribution assessable without treating document reorganization as scientific closure.

The completed evidence for this item is in Table II; the exact row files and hashes are retained in the current response map.

R120 source route. The exact current file routes are `CONTENT_RETENTION_MAP120.md`, `REVISION_BOUNDARY120.md`, `provenance/HISTORICAL_ROUTE_BINDINGS120.json`.

Scientific boundary. A shorter article must not lose the neural continuation/actor construction or conceal negative findings. Source identity, proof validity, numerical execution and editorial assessment remain separate claims.

2.21 Minor 1. Error-versus-information terminology

Original advisory request.

> 1. State 'N^{-2}' consistently as an error-versus-information rate; "information exponent two" is easy to misread.

Completed inherited response. The current theorem and response use error-versus-information rate for the N^{-2} width order. They distinguish risk width, constrained action gap, actual regret and observed work counts and state the separate original-investment action-transfer result.

Concrete R120 author action. The new result is called interval width versus scalar point information, with order $\min\{KR(1-\alpha)N^{-2}, ER(1-\alpha)N^{-1}\}$. Original-policy contrast, certified gain, regret, value width, squared action-distance bound and complete work are separate quantities throughout the proofs and all 64 objective/cost cells. The independently computed optimum is an enclosure, so the displayed RMS quantities are upper bounds rather than measured errors to a known exact optimizer. The achieved R119 N^{-2} tail-uniform AVaR width result retains its established attribution; the executed economic gains require their own policy evidence.

The completed evidence for this item is in Table II; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition S.8, Proposition S.12. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `INDEPENDENT_PROTOCOL120.md`, `code120/independent120.py`.

Scientific boundary. A rate for one of these quantities is not silently advertised as a rate for another.

2.22 Minor 2. Oracle qualifiers

Original advisory request.

> 2. Put "deterministic, point-value, univariate, fixed-alpha" adjacent to every high-level summary of the lower bound.

Completed inherited response. High-level summaries state deterministic exact point values, univariate uniform risk support and a known convex Lipschitz class. The new result improves the historical fixed-alpha claim to alpha-uniform constants for N at least 32, while retaining the limits concerning bit cost, formula access and multiple innovations.

Concrete R120 author action. Deterministic, exact scalar point-value, univariate, uniform-law, convex-Lipschitz and fixed-known-tail qualifiers appear with the new information statement. Purchasing the two continuation values at a point costs two queries. The original smooth investment formula is a narrower class than the lower adversary, and risk-tail localization, finite point precision, terminal solves and the future controller remain separately paid. R119's risk-level-uniform width theorem is retained under its own assumptions. The completed two-risk experiment does not establish randomized, multivariate or arbitrary-law oracle optimality.

The completed evidence for this item is in Table B.III; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition S.8. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`.

Scientific boundary. The fixed-alpha limitation of the older upper bound has been improved in R119; randomized, multivariate and arbitrary-law claims have not followed automatically.

2.23 Minor 3. Tail dependence

Original advisory request.

> 3. Discuss the opposite tail-level behavior of the lower and upper constants as 'alpha' approaches one.

Completed inherited response. The mismatch is addressed mathematically: paid tail localization and selected-tail quadrature give an upper width proportional to KR(1-alpha)N^{-2}, matching the retained lower order in both information and tail mass. The proof handles alpha=0 separately, excludes alpha=1 in normalized tails, and charges sign queries.

Concrete R120 author action. The achieved R119 order $KR(1-\alpha)N^{-2}$ is preserved. R120's fixed-tail comparison intersects widths bounded by $2KR(1-\alpha)/n^2$ and $ER(1-\alpha)/n$, with the original smooth coefficient supplied separately. The action modulus includes its explicit $\alpha/(1-\alpha)$ dependence; this prevents a small tail mass alone from implying a cheap original-model computation near $\alpha = 1$. The full-support $\alpha = 0$ boundary is separate and the normalized-tail statement excludes $\alpha = 1$. Actual new observations use only the predeclared $1/2$ and $3/4$ risk levels and retain every paid tail/terminal operation.

The completed evidence for this item is in Table III, Table II; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition 11, Proposition S.8, Theorem S.9, Proposition S.10, Proposition S.11. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`.

Scientific boundary. Do not relabel the established R119 tail-uniform result as an R120 discovery or infer negligible implementation cost near alpha=1.

2.24 Minor 4. Acquisition restrictions

Original advisory request.

> 4. The readout coefficient bounds are acquisition restrictions, not characterizations of the best approximation; report sensitivity.

Completed inherited response. Coefficient signs, bounds, dictionary size, rounding and feature selection remain acquisition restrictions, not a characterization of best approximation in a representation family. The fixed current models preserve their restrictions and exact source identities.

Concrete R120 author action. The actual actor artifacts and acquisition table preserve all prescribed feature counts, coefficient signs/bounds, rounding and ties. The constrained rank-six readouts are compared with the larger fixed polynomial and grid controls under the same 256 original-V_1 label sites per risk. ReLU and ordinary hinge have identical retained parameters and actions, with the full shared-fit charge assigned to each standalone deployment. These are acquired-model comparisons under known restrictions. They are not best-approximation results for unrestricted representation classes, and the experiment supplies no post-outcome coefficient or architecture sensitivity menu.

The completed evidence for this item is in Table II, Table III; the exact row files and hashes are retained in the current response map.

R120 source route. The exact current file routes are `PROTOCOL120.md`, `code120/campaign120.py`, `code120/improvement120.py`.

Scientific boundary. The inherited constrained readout does not characterize best approximation in the unrestricted function class. A broad sensitivity study remains unexecuted unless separately supplied.

2.25 Minor 5. Residual span and action gap

Original advisory request.

> 5. Distinguish validated residual span from Bellman action gap in terminology and tables.

Completed inherited response. The current text and tables distinguish validated residual span, quadrature width, tail-selection defect, original Bellman action gap, regret bound and signed actual-policy difference. The action certificate uses original supports and the implemented actor allowance rather than substituting a fit norm for decision quality.

Concrete R120 author action. Every primary group now has separate original-policy bounds, independent original-optimum bounds, mean regret, RMS action-error upper bound and own-incumbent gain, alongside production contrast and certificate counters. The independent validator supplies 55,104 retained all-action support rows and all 4,096 policy comparisons. For example, the ReLU P gain is economically positive against its incumbent while its mean difference from L is exactly zero. A narrower value interval, smaller fit residual or lower regret ceiling is therefore never counted as an observed policy-value decrease by itself. The 726 versus 3,370 containment directions also retain their different independent-verification implications.

The completed evidence for this item is in Table II; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition S.8, Proposition S.12. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `INDEPENDENT_PROTOCOL120.md`, `code120/independent120.py`.

Scientific boundary. An improved fit statistic, tighter interval or lower regret ceiling alone is not an observed decrease in the policy's true economic cost.

2.26 Minor 6. Rejections and unsuccessful work

Original advisory request.

> 6. Report rejection changes beside every span improvement.

Completed inherited response. The adverse R97 span result is retained beside its rejection outcome: all 36 candidates lower span while total rejection rises by 113, from 644 to 757. R119 A/B/C report all 244 distinct repaired primary proposals and zero within-arm policy changes; better certification work is not called improved policy value.

Concrete R120 author action. Every attempted candidate and resolution, rejection, exhausted ladder, unchanged action and fallback is retained in the actual traces and complete tables. P's 2,257 refusals and D's 633 refusals appear beside accepted-step counts, as do 785 and 194 exhausted ladders. Mixed-bound use is reported at all 9,016 paid levels: 5,961 active upper bounds and two necessary accepted records for one economic candidate. The transferred incumbent bound is inactive in all 2,290 density pairs. Its mathematical validity is not converted into an unobserved reduction of already purchased tail endpoints, candidate-upper work or total time. Every unresolved and adverse comparison remains included.

The completed evidence for this item is in Table B.III; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Proposition 11, Theorem S.9, Proposition S.10, Proposition S.11. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`, `code120/campaign120.py`.

Scientific boundary. No R120 summary may select only positive gains or omit an unresolved or failed comparison.

2.27 Minor 7. Observed engineering ranges

Original advisory request.

> 7. Label the acquisition bracket as an observed engineering range, not a confidence interval.

Completed inherited response. Acquisition and timing ranges are descriptive engineering observations, with all repetitions and scopes retained. C service medians exclude separately measured loading and startup; the full 144-state component account includes its actual acquisition and primary load. No observed clock range is called a confidence interval.

Concrete R120 author action. The three process-specific measurements, all 192 cohort clocks and their medians and ranges are labeled descriptive engineering observations. All 6,144 comparisons of actions, intervals and query counts match, so the reported conditional pairing of timing and economic outcomes is actually checked. These clocks remain distinct from outward rational value enclosures and from the explicitly proved simultaneous 95% population family of 48 groups and 192 events. Timing replicates do not increase its 64 independent initial-state units. The inherited seven R119 clocks retain their separate episode and do not enlarge the current sample.

The completed evidence for this item is in Table III, Table IV; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Proposition 12. The exact current file routes are `PROTOCOL120.md`, `code120/campaign120.py`, `code120/improvement120.py`, `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `INDEPENDENT_PROTOCOL120.md`, `code120/stats120.py`.

Scientific boundary. The three R120 timing processes and seven inherited R119 process clocks do not by themselves establish a population or deployment confidence interval.

2.28 Minor 8. Adoption over the conventional frontier

Original advisory request.

> 8. State visibly that there are zero adoption regions over quadratic least squares.

Completed inherited response. The original R97 1,800-cell grid retains zero learned-service adoption regions against quadratic least squares. The new C table also retains the stronger quadratic full cost and mean-policy value versus ReLU at both risk levels. C improves the common certification service but does not reverse the neural adoption finding.

Concrete R120 author action. The old zero-adoption-over-quadratic result and the R119 conventional frontier remain explicit. In the new fixed comparison, positive gains against learned incumbents coexist with an online advantage against all 14 learned completed means and exact ReLU–hinge identity. All 32 ReLU intersections against the predeclared full conventional menu, across four amortization volumes and three clocks, have empty sufficient-price sets and empty intersections of upper not-ruled-out sets. Among individual clock comparisons for ReLU, 12 of 96 upper regions remain nonempty; two of eight actual-primary workload upper regions are also nonempty. These are unresolved comparisons, retained separately from the three-clock intersections. The 17 nonempty sufficient menu regions across representations belong to completed conventional policies. The 112 separate actual-64-state price domains retain the actual workload and charges. These strict recorded-menu conditions and own-incumbent gains do not assert population adoption or rule on every possible deployment criterion.

The completed evidence for this item is in Table B.VIII, Table II; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Proposition 12. The exact current file routes are `PROTOCOL120.md`, `code120/campaign120.py`, `code120/improvement120.py`, `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `INDEPENDENT_PROTOCOL120.md`, `code120/stats120.py`.

Scientific boundary. A gain over a learned arm's own incumbent or a conditional symbolic price region does not erase those historical findings.

2.29 Minor 9. Verified approximation and HJB literature

Original advisory request.

> 9. Compare the certification contribution carefully with existing verified approximation and HJB-residual literatures.

Completed inherited response. The introduction identifies classical convex integration, AVaR duality, approximate policy improvement and verified residual ideas, then isolates the paid tail localization, matched tail-information constants and original-action certificate. The foundations state the stopped BSDE/HJB residual and boundary hypotheses explicitly.

Concrete R120 author action. The main text locates the mixed integration interface relative to retained verified approximation, Bellman/HJB residual ideas, constrained safeguards and the source-owned common-risk identity. Complete proofs retain the original filtration, boundaries and one-sided future-policy allowance. Exact binomial inversion and bounded-mean concentration are attributed as classical ingredients; the new population conclusion is the specified transfer from actual certified gains. The stopped Brownian verification subcase retains its hypotheses and complete boundary argument. The present two-date validation supplies no proof or experimental substitute for a general coupled NDU/FBSDE, nonlinear recursive-preference application, temporal-self equilibrium or game.

The completed evidence for this item is in Table IV; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition 11, Proposition 12, Proposition S.8, Theorem S.9, Proposition S.10, Proposition S.11. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`, `INDEPENDENT_PROTOCOL120.md`, `code120/stats120.py`.

Scientific boundary. Citations do not replace a proof; the stopped Brownian subcase does not imply a general coupled NDU/FBSDE or game result.

2.30 Minor 10. Title and neural content

Original advisory request.

> 10. The title should not carry more neural-specific content than the theorem and evidence identify.

Completed inherited response. Neural Bellman Operators remains the central author research identity. The title and main text retain learned continuations, recursive operators and actor/critic construction while explicitly identifying the new information and certification results as shared by equivalent neural and conventional representations.

Concrete R120 author action. The title and active main text retain Neural Bellman Operators as the study of learned original continuations, recursive operators and feasible actor/critic decisions. The new action-dependent comparison and safeguard are attributed to an eligible-method interface shared by the neural and conventional representations. Their original-policy gains are actually measured and independently bounded; conventional hinge remains identical to ReLU, and the strongest original-formula controls remain visible. This establishes a concrete shared numerical-methods result inside the NBO problem while leaving equal-information neural advantage and action-aware learning as explicit ongoing research requirements.

The completed evidence for this item is in Table II; the exact row files and hashes are retained in the current response map.

R120 source route. The written mathematical route is Theorem 9, Theorem 10, Proposition S.8. The exact current file routes are `new120/policy_improvement_main.tex`, `new120/policy_improvement_proofs.tex`, `code120/improvement120.py`, `PROTOCOL120.md`, `code120/campaign120.py`.

Scientific boundary. The title does not license attributing conventional hinge geometry, shared original-value information or a common safeguard to neural networks alone.

3 All earlier reports and independently frozen work

The retained major and additional response maps preserve 620 major blocks, 394 additional requests and 35 overlapping parent treatments. These are 1,014 routed request records and 1,049 original item/parent routes, not 1,049 independent scientific issues. All 1,146 hierarchical report sections retain their classification. The new `provenance/HISTORICAL_ROUTE_BINDINGS120.json` points every original request to its exact R119 author treatment, remaining requirement and source binding. No new report on R120 is invented, and no blanket scientific closure is assigned.

The original Bellman objective and information filtration, own-future continuation construction, feasible actor, complete-policy cost, boundary conditions and actor/critic graph remain central. R119's finite-horizon existence and selector proofs, constrained envelope, exact observation transfer, original-value supports and stopped Brownian BSDE verification remain active under their stated assumptions. Corrected composite-loss, viscosity, portfolio, recursive-utility and source-constructed-curve errors remain visible in the development companion. General coupled NDU/FBSDEs, temporal-self equilibria and games do not inherit a discrete investment experiment's validity or numerical ranking.

R98 retains its seven-candidate acquisition menu, its original source and actor-selection barriers, and its own independent-test protocol. R115 retains its fixed actors, exact pre-evaluation commitment to 8,192 random bytes and 1,024 dyadic states, and its unchanged R114 population protocol with simultaneous inferential allocations. The sample-barrier statement records its own time; it does not establish that a historical process is still running now. Neither study is completed by the new R120 sample or by borrowing another episode's observations. R111–R114 and R116–R118 likewise retain their declared sources, model boundaries, amendments, failures and incomplete handoffs. New work on a related idea must credit its conceptual lineage and identify its own implementation and evidence.

The inherited R119 A/B/C source freezes, D's separate known-catalogue validation, the unchanged-source R117/R118 reproductions and the later publication/custody receipts remain distinct. New policy keys require new value evidence. An old shared-compiler reference or a broader independent interval cannot be relabeled a verification of newly returned actions or of every finer old endpoint. The complete history remains available in the original branches and the source-bound retention maps.

4 Evidence and source/PDF publication

The primary and three timing processes completed with an external exit-zero receipt at 08:09:59 UTC on 19 October 2026. Independent validation completed at 08:42:24 UTC; the fixed statistics completed at 08:43:06 UTC. The first read-only reporter completed at 08:44:13 UTC and binds 777 unchanged input files and 36 stable output files. The original protocol and source F, actual actors M, single sample D and pre-outcome reporting specification R remain immutable. The primary evidence checkpoint is separately published as `a6f3b8278d4502b174b05b19fafd5aacabaa60a3`. All exact launches, stdout, process receipts, primitive traces, refusals, timing repetitions and output tables are retained. The reporter is a read-only reaggregation of actual records, not a second economic experiment.

The independent evaluator supplies original-policy and original-optimum enclosures from separately implemented rational primitives. Its 1,806.153-second inner process is charged separately from deployment, and its wider enclosure does not validate every finer production endpoint. The post-timing algebra diagnostic preserves its first derivative source-path failure and the corrected second attempt's 17,039 checks over 1,333 cases; neither changes the frozen scientific source.

`EVIDENCE_MANIFEST120.json` binds the complete current evidence, archive transport, executed reporter products and source requirements. The bounded verifier checks bytes, complete censuses and recorded rational invariants, with separately identified statistics/report replays. It does not claim to rerun acquisition, independent economic primitives or timings. The source/PDF publication uses separate `SOURCE_PDF_MANIFEST120.json` and `PUBLICATION120.json` declarations for the actual author-source commit, four PDFs, build receipts and dated remote CI observations. Those later derivative receipts avoid an author-source hash self-reference and preserve earlier

status observations. No queued or future job, inherited R119 success or merely prepared workflow is asserted by this source as an R120 CI success. The original NBO/Econometrica research goal remains active with the proved and executed advances, adverse costs, unresolved incremental superiority and explicit remaining neural-learning obligations.